Exercise 3 — Concepts

Advanced System Theory · Week 03: Response of Linear Systems

Course: Advanced System Theory (Paderborn University, Automatic Control Group RAT, Prof. Dr. Erdal Kayacan), winter semester 2026/27 **Lecture:** Week 03 — Response of linear systems **Covers:** everything needed for Exercise 3 — Task 1 (diagonalization, modes, impulse response), Task 2 (recovering $\boldsymbol{A}$ and $\boldsymbol{\Phi}$ from trajectories), Task 3 (choosing initial conditions), Task 4 (Cayley–Hamilton, response to a delayed step)

This sheet starts with the **intuition** — what a system response is and why we need it — then builds every tool from the ground up. Each method is given as a recipe with a worked example. The full solutions are in `solutions.md`; the scripts in `python/` check every number.

Notation (as in the lecture slides). Bold symbols are vectors and matrices: state $\boldsymbol{x}(t) \in \mathbb{R}^n$, input $\boldsymbol{u}(t)$, output $\boldsymbol{y}(t)$, system matrices $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$, forcing term $\boldsymbol{r}(t)$. $\boldsymbol{\Phi}(t, t_0)$ is the state transition matrix, $\exp(\boldsymbol{A}t) = e^{\boldsymbol{A}t}$ the matrix exponential, $\boldsymbol{P}$ the eigenvector matrix and $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, $\boldsymbol{p}_i$ the i-th column of $\boldsymbol{P}$ and $\tilde{\boldsymbol{p}}_i$ the i-th row of $\boldsymbol{P}^{-1}$, $\alpha_i(t)$ the Cayley–Hamilton coefficients, $\sigma(t)$ the unit step, $\delta(t)$ the Dirac impulse, $\boldsymbol{H}(t)$ the impulse response matrix. Subscripts h and p mark the homogeneous and particular solution. $j = \sqrt{-1}$.

Contents

0. The big picture: what this lecture is about
1. Warm-up: the scalar case $\dot{x} = ax + bu$
2. Mathematical toolkit
3. Homogeneous and inhomogeneous equations
4. The state transition matrix
5. The matrix exponential
6. Method 1 — series expansion and nilpotent matrices
7. Method 2 — similarity transform (diagonalization)
8. Modes
9. Method 3 — Jordan form (non-diagonalizable A)
10. Method 4 — Cayley–Hamilton theorem
11. Which method to choose?
12. Response with an input: the convolution integral
13. Standard test signals, impulse and step response
14. Worked lecture example: total response
15. Two “inverse” problems from the exercise
16. Cheat sheet

---

0. The big picture: what this lecture is about

0.1 What is a "system response"?

In Week 2 we learned to *write down* a system: $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}$, $\boldsymbol{y} = \boldsymbol{C}\boldsymbol{x} + \boldsymbol{D}\boldsymbol{u}$. That equation only says **how fast the state is changing right now**. It does not directly say **where the state will be in 5 seconds**.

The **system response** answers exactly that question:

> *Given where the system starts ($\boldsymbol{x}_0$ at time t_0) and what we do to it (the input $\boldsymbol{u}(t)$), what are the state $\boldsymbol{x}(t)$ and the output $\boldsymbol{y}(t)$ at every later time?*

In other words: **solving the differential equation**. The state equation is the *rule*; the response is the *movie* that the rule produces.

An everyday picture: a car. The rule is "acceleration = (engine force − friction)/mass". The response is the actual speed-versus-time curve when you start at 30 km/h and press the pedal halfway for 10 seconds.

The response always consists of **two parts** (thanks to the superposition principle of Week 2, Definition 1.1):

Part	Also called	Cause	Car picture
$\boldsymbol{x}_h$, homogeneous solution	**zero-input / natural / free** response	the initial state alone ($\boldsymbol{u} = 0$)	you take your foot off the pedal: the car rolls out from 30 km/h
$\boldsymbol{x}_p$, particular solution	**zero-state / forced** response	the input alone ($\boldsymbol{x}_0 = 0$)	starting from standstill, you press the pedal

$$\text{total response} = \text{natural response} + \text{forced response}.$$

Because the system is linear, the two can be computed **separately** and simply **added**.

0.2 Why do we need it?

- **Prediction and simulation.** Knowing $\boldsymbol{x}(t)$ tells you whether a robot arm reaches its target, whether a tank overflows, how long a motor needs to reach speed.
- **Performance.** Rise time, overshoot, oscillation, settling time — all are read off the response to a standard test signal (step, impulse).
- **Stability.** Does the response die out, stay bounded or blow up? The answer sits in the exponentials $e^{\lambda_i t}$ of the response (formalised in Weeks 9–10).
- **Understanding.** The response decomposes into **modes** — the natural "building blocks" of motion the system can perform (Section 8). Knowing them tells you which initial states or inputs excite slow, fast, oscillating or unstable behaviour.
- **Foundation for the rest of the course.** The state transition matrix $e^{\boldsymbol{A}t}$ reappears in discretization (Week 4: $\boldsymbol{A}_d = e^{\boldsymbol{A}T}$), in the Laplace transform (Week 5: $\mathcal{L}\{e^{\boldsymbol{A}t}\} = (s\boldsymbol{I} - \boldsymbol{A})^{-1}$), in stability (Weeks 9–10), and in controllability/observability (Weeks 11–12).

The key promise of this lecture (slide 43): for linear time-invariant (LTI) systems a **general closed-form solution always exists**. For nonlinear systems it usually does not — which is why Week 2 linearised them first.

0.3 The lecture in one picture

$$\boldsymbol{y}(t) = \underbrace{\boldsymbol{C}\, e^{\boldsymbol{A}(t-t_0)}\boldsymbol{x}_0}_{\text{natural response}} + \underbrace{\boldsymbol{C}\int_{t_0}^{t} e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\,\boldsymbol{u}(\tau)\,\mathrm{d}\tau + \boldsymbol{D}\boldsymbol{u}(t)}_{\text{forced response}}$$

Everything in the lecture serves this one formula:

Topic	Intuition in one line	Section
Homogeneous vs. inhomogeneous ODE	"no input" vs. "with input" — superposition lets us treat them separately	3
State transition matrix $\boldsymbol{\Phi}(t, t_0)$	a **time machine** for the state: multiply the state at t_0 by $\boldsymbol{\Phi}$ and you get the state at t	4
Matrix exponential $e^{\boldsymbol{A}t}$	the matrix version of e^{at} from school — for LTI systems it *is* $\boldsymbol{\Phi}$	5
Series / nilpotent	add up $\boldsymbol{I} + \boldsymbol{A}t + \boldsymbol{A}^2t^2/2 + \ldots$; sometimes it stops after a few terms	6
Similarity transform	change coordinates so that the states **decouple** — then each one is a scalar $e^{\lambda t}$	7
Modes	the response is a **sum of simple motions** $e^{\lambda_i t}$ in fixed directions $\boldsymbol{p}_i$ — like the notes in a chord	8
Jordan form	if there are not enough eigenvectors, terms like $t\,e^{\lambda t}$ appear	9
Cayley–Hamilton	any function of an $n \times n$ matrix is a polynomial of degree $n-1$ — only n unknown coefficients	10
Convolution integral	the forced response **adds up the after-effects of all past inputs**, each weighted by how long ago it happened	12
Impulse response $\boldsymbol{H}(t)$	the system's **fingerprint**: hit it once, record the output; every other response follows from it	13

0.4 Which section is needed for which task?

Exercise task	Needs
Task 1(a) diagonalization of $e^{\boldsymbol{A}t}$	§2.1 complex numbers, §5, §7 (incl. §7.4 complex eigenvalues)
Task 1(b) modes, divergence	§8
Task 1(c) impulse response	§2.4 Dirac impulse, §12, §13.2

Exercise task	Needs
Task 2 recover $\boldsymbol{A}$ and $\boldsymbol{\Phi}$	§4.3, §5.2, §15.1
Task 3 initial condition for $y(t) = te^{-t}$	§2.3 linear independence, §9, §15.2
Task 4(a) Cayley–Hamilton	§10
Task 4(b) response, step at $t_0 > 0$	§4.2 (backwards in time), §12, §13.3

---

1. Warm-up: the scalar case $\dot{x} = ax + bu$

Everything in this lecture is the matrix version of what happens with **one** state. Understanding the scalar case first makes the rest feel natural.

1.1 No input: $\dot{x} = ax$

Try $x(t) = e^{a(t-t_0)}x_0$. Its derivative is $a\,e^{a(t-t_0)}x_0 = a\,x(t)$ ✓, and at $t = t_0$ it gives $e^0 x_0 = x_0$ ✓. So

$$x(t) = e^{a(t-t_0)}\,x_0.$$

- $a < 0$: the state decays — the system "forgets" its initial condition.
- $a > 0$: the state grows — unstable.
- $|a|$ sets the **speed**; $1/|a|$ is the **time constant**.

The factor $e^{a(t-t_0)}$ is the scalar "state transition": it carries the state from time t_0 to time t.

1.2 With input: integrating factor

$\dot{x} - ax = bu$. Multiply by e^{-at}; the left side becomes a perfect derivative:

$$e^{-at}\dot{x} - ae^{-at}x = \frac{\mathrm{d}}{\mathrm{d}t}\big(e^{-at}x\big) = e^{-at}\,b\,u(t).$$

Integrate from t_0 to t:

$$e^{-at}x(t) - e^{-at_0}x_0 = \int_{t_0}^{t} e^{-a\tau} b\,u(\tau)\,\mathrm{d}\tau \quad\Longrightarrow\quad x(t) = \underbrace{e^{a(t-t_0)}x_0}_{\text{natural}} + \underbrace{\int_{t_0}^{t} e^{a(t-\tau)}\,b\,u(\tau)\,\mathrm{d}\tau}_{\text{forced}}.$$

Reading the integral. The input at an earlier time τ gives the state a little "kick" $b\,u(\tau)\,\mathrm{d}\tau$. By time t that kick has evolved for $t - \tau$ seconds, so it has been multiplied by $e^{a(t-\tau)}$. The integral **adds up all these past kicks**. For $a < 0$, old kicks ($t - \tau$ large) have faded — the system has a **fading memory**.

1.3 Step input

$u(t) = 1$ for $t \geq t_0$, $x_0 = 0$:

$$x(t) = b\int_{t_0}^{t} e^{a(t-\tau)}\,\mathrm{d}\tau = \frac{b}{a}\big(e^{a(t-t_0)} - 1\big) \xrightarrow{a<0,\ t\to\infty} -\frac{b}{a}.$$

The matrix version of exactly this formula is used in Task 4 (§13.3).

The rest of the lecture replaces the number a by the matrix $\boldsymbol{A}$. The only real difficulty is: *what is $e^{\boldsymbol{A}t}$, and how do we compute it?*

---

2. Mathematical toolkit

2.1 Complex numbers and Euler's formula (needed for Task 1)

A complex number is $z = \sigma + j\omega$ with real part σ, imaginary part ω and $j^2 = -1$. Its **conjugate** is $\bar{z} = \sigma - j\omega$.

- Multiplication: $(a + jb)(c + jd) = (ac - bd) + j(ad + bc)$.
- Division: multiply numerator and denominator by the conjugate of the denominator, e.g. $\dfrac{1}{a+jb} = \dfrac{a-jb}{a^2+b^2}$.
- $z + \bar{z} = 2\,\mathrm{Re}\,z$ (real), $z - \bar{z} = 2j\,\mathrm{Im}\,z$.

Euler's formula:

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t), \qquad \cos(\omega t) = \frac{e^{j\omega t} + e^{-j\omega t}}{2}, \qquad \sin(\omega t) = \frac{e^{j\omega t} - e^{-j\omega t}}{2j}.$$

So for a complex eigenvalue $\lambda = \sigma + j\omega$:

$$e^{\lambda t} = e^{\sigma t}e^{j\omega t} = e^{\sigma t}\big(\cos\omega t + j\sin\omega t\big), \qquad |e^{\lambda t}| = e^{\sigma t}.$$

Interpretation: the **real part σ decides growth or decay** (envelope $e^{\sigma t}$), the **imaginary part ω decides the oscillation frequency**. A real matrix always has complex eigenvalues in **conjugate pairs** $\sigma \pm j\omega$, and their contributions combine into real cos/sin terms.

Roots of $\lambda^2 + a_1\lambda + a_0 = 0$ with negative discriminant: $\lambda = -\dfrac{a_1}{2} \pm j\sqrt{a_0 - \dfrac{a_1^2}{4}}$.

2.2 Matrix facts

- **2×2 inverse:** $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dfrac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (works for complex entries too).
- **Eigenvalues:** roots of $\det(\lambda\boldsymbol{I} - \boldsymbol{A}) = 0$; for 2×2, $\lambda^2 - \mathrm{tr}(\boldsymbol{A})\lambda + \det(\boldsymbol{A}) = 0$.
- **Eigenvectors:** solve $(\lambda_i\boldsymbol{I} - \boldsymbol{A})\boldsymbol{p} = 0$; any nonzero multiple is also an eigenvector.
- **Triangular matrices:** eigenvalues are the diagonal entries.
- **Block-diagonal matrices** $\mathrm{diag}(\boldsymbol{A}_1, \boldsymbol{A}_2)$: everything (eigenvalues, powers, exponentials) can be done block by block.
- **Matrix product order matters:** $\boldsymbol{AB} \neq \boldsymbol{BA}$ in general.

2.3 Linear independence of exponential functions (needed for Tasks 2 and 3)

The functions $e^{\lambda_1 t}, e^{\lambda_2 t}, \ldots$ (distinct λ_i), together with $t\,e^{\lambda t}, t^2e^{\lambda t}, \ldots$, are **linearly independent**: if

$$c_1e^{\lambda_1 t} + c_2\,t\,e^{\lambda_1 t} + c_3e^{\lambda_2 t} + \cdots = 0 \quad \text{for all } t,$$

then **every coefficient is zero**. Consequence — **coefficient matching**: two such sums are equal for all t only if the coefficients of each function agree. E.g. $(a+b)e^{-t} + b\,te^{-t} + ce^{2t} = te^{-t}$ for all t forces $a+b=0$, $b=1$, $c=0$.

2.4 The unit step and the Dirac impulse

- **Unit step:** $\sigma(t) = 1$ for $t \geq 0$, 0 otherwise. A delayed step $\sigma(t-t_0)$ switches on at t_0.
- **Rectangle:** $\Pi(t) = 1/\epsilon$ for $0 \leq t \leq \epsilon$, 0 otherwise — area always 1.
- **Dirac impulse:** $\delta(t) = \lim_{\epsilon\to 0} \Pi(t)$ — infinitely short, infinitely high, **area 1**. It is not an ordinary function; it is defined by what it does inside an integral, the **sifting property**:

$$\int_a^b f(\tau)\,\delta(\tau - t_1)\,\mathrm{d}\tau = f(t_1) \qquad \text{if } a < t_1 < b.$$

The impulse "picks out" the value of f at the instant of the impulse. Physical picture: a hammer blow — a huge force for a very short time, delivering a finite momentum.

Convention for an impulse at $t = 0$: the impulse acts at the very beginning, so the integral $\int_0^t$ is understood as starting just before it (0^-) and the whole area counts: $\int_{0^-}^t f(\tau)\delta(\tau)\,\mathrm{d}\tau = f(0)$ for $t > 0$.

2.5 Integrating exponentials

$\int e^{c\tau}\,\mathrm{d}\tau = \frac{1}{c}e^{c\tau}$ ($c \neq 0$), and therefore $\int_{t_0}^t e^{a(t-\tau)}\,\mathrm{d}\tau = \frac{1}{a}\big(e^{a(t-t_0)} - 1\big)$. For matrices: §13.3.

---

3. Homogeneous and inhomogeneous equations

3.1 Definition 1.1

We consider the system of n linear **homogeneous** ODEs

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}(t)\boldsymbol{x}(t)$$

and the system of n linear **inhomogeneous** ODEs

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}(t)\boldsymbol{x}(t) + \boldsymbol{r}(t),$$

where $\boldsymbol{A}(t)$ is the dynamic matrix, $\boldsymbol{r}(t) \in \mathbb{R}^n$ the system input (forcing term) and $\boldsymbol{x}(t) \in \mathbb{R}^n$ the state. For specified initial conditions $\boldsymbol{x}(t_0) = \boldsymbol{x}_0$ these systems possess **unique** solutions for every $(t_0, \boldsymbol{x}_0)$.

For a state-space model the forcing term is $\boldsymbol{r}(t) = \boldsymbol{B}\boldsymbol{u}(t)$ (slide 33).

3.2 Structure of the solution (slide 6)

$$\boldsymbol{x}(t, t_0, \boldsymbol{x}_0) = \underbrace{\boldsymbol{\Phi}(t, t_0)\boldsymbol{x}_0}_{\boldsymbol{x}_h} + \underbrace{\int_{t_0}^t \boldsymbol{\Phi}(t,\tau)\boldsymbol{r}(\tau)\,\mathrm{d}\tau}_{\boldsymbol{x}_p}.$$

- **Homogeneous solution x_h:** nonzero initial conditions, zero forcing ⇒ **zero-input response**.
- **Particular solution x_p:** zero initial conditions, nonzero forcing ⇒ **zero-state response**.
- For given $\boldsymbol{x}_0$ and given forcing, $\boldsymbol{x}(t)$ is known for **all** t. *This is why we call $\boldsymbol{x}$ the state:* it contains everything about the past that matters for the future.

---

4. The state transition matrix

4.1 Definition 1.2

The **state transition matrix** $\boldsymbol{\Phi}(t, t_0)$ is the unique (nonsingular) solution of the matrix equation

$$\dot{\boldsymbol{\Phi}}(t, t_0) = \boldsymbol{A}(t)\boldsymbol{\Phi}(t, t_0) \qquad \text{with} \qquad \boldsymbol{\Phi}(t_0, t_0) = \boldsymbol{I}.$$

It is uniquely determined by $\boldsymbol{A}(t)$, and the solution of the homogeneous ODE is

$$\boldsymbol{x}(t, t_0, \boldsymbol{x}_0) = \boldsymbol{\Phi}(t, t_0)\, \boldsymbol{x}_0 \qquad \forall t.$$

Intuition — a time machine for the state. Give $\boldsymbol{\Phi}(t, t_0)$ the state at time t_0 and it returns the state at time t. *Why does $\boldsymbol{\Phi}$ satisfy the same ODE as $\boldsymbol{x}$?* Because each **column** of $\boldsymbol{\Phi}$ is itself a solution: the i-th column is the trajectory that starts at the i-th unit vector $\boldsymbol{e}_i$ (since $\boldsymbol{\Phi}(t_0, t_0)\boldsymbol{e}_i = \boldsymbol{e}_i$). Any other initial state is a combination $\boldsymbol{x}_0 = \sum x_{0,i}\boldsymbol{e}_i$, so by superposition its trajectory is the same combination of the columns — that is $\boldsymbol{\Phi}\boldsymbol{x}_0$.

4.2 Properties (slide 8)

For any $t, \tau, \sigma \in \mathbb{R}$:

1. **Composition:** $\boldsymbol{\Phi}(t, \tau) = \boldsymbol{\Phi}(t, \sigma)\boldsymbol{\Phi}(\sigma, \tau)$ — travelling from τ to σ and then from σ to t is the same as travelling from τ to t directly.
2. **Invertibility:** $\det \boldsymbol{\Phi}(t, t_0) \neq 0$ and

   $$\big(\boldsymbol{\Phi}(t, t_0)\big)^{-1} = \boldsymbol{\Phi}(t_0, t).$$

   The inverse travels **backwards**. In particular, $\boldsymbol{\Phi}(t, t_0)$ with $t < t_0$ is perfectly meaningful: if you know the state at t_0, you can compute the state at **earlier** times. *(Needed in Task 4, where the state is given at $t_0 > 0$ and the response is asked for $0 \leq t \leq t_0$.)*
3. Closed-form expressions of $\boldsymbol{\Phi}(t, t_0)$ exist only for special classes of $\boldsymbol{A}(t)$ — most importantly for **constant** $\boldsymbol{A}$ (next section).

4.3 Building $\boldsymbol{\Phi}$ from known trajectories (needed for Task 2)

Suppose you know n trajectories $\boldsymbol{x}^{(1)}(t), \ldots, \boldsymbol{x}^{(n)}(t)$ of $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}$, starting from **linearly independent** initial states $\boldsymbol{x}^{(i)}(0)$. Each satisfies $\boldsymbol{x}^{(i)}(t) = \boldsymbol{\Phi}(t, 0)\boldsymbol{x}^{(i)}(0)$. Stack them as columns:

$$\underbrace{\big[\boldsymbol{x}^{(1)}(t) \;\cdots\; \boldsymbol{x}^{(n)}(t)\big]}_{\boldsymbol{X}(t)} = \boldsymbol{\Phi}(t, 0)\, \underbrace{\big[\boldsymbol{x}^{(1)}(0) \;\cdots\; \boldsymbol{x}^{(n)}(0)\big]}_{\boldsymbol{X}_0} \quad \Longrightarrow \quad \boxed{\boldsymbol{\Phi}(t, 0) = \boldsymbol{X}(t)\, \boldsymbol{X}_0^{-1}}$$

$\boldsymbol{X}_0$ is invertible because its columns are independent. ($\boldsymbol{X}(t)$ is called a **fundamental matrix**.) Then $\boldsymbol{A}$ follows from Definition 1.2 at $t = 0$: $\dot{\boldsymbol{\Phi}}(0, 0) = \boldsymbol{A}\boldsymbol{\Phi}(0, 0) = \boldsymbol{A}$, i.e.

$$\boxed{\boldsymbol{A} = \dot{\boldsymbol{\Phi}}(t,0)\big|_{t=0}.}$$

---

5. The matrix exponential

5.1 Theorem 1.1 — state transition matrix for autonomous LTI systems

For time-invariant $\boldsymbol{A}$, the state transition matrix is

$$\boldsymbol{\Phi}(t,t_0) = \exp\big[\boldsymbol{A}(t-t_0)\big], \qquad \text{in particular} \qquad \boldsymbol{\Phi}(t,0) = \exp(\boldsymbol{A}t) = \sum_{k=0}^{\infty} \frac{t^k}{k!}\boldsymbol{A}^k = \boldsymbol{I} + \boldsymbol{A}t + \frac{\boldsymbol{A}^2t^2}{2!} + \frac{\boldsymbol{A}^3t^3}{3!} + \cdots$$

This is the same power series as the scalar $e^{at} = 1 + at + \frac{a^2t^2}{2!} + \cdots$, with a replaced by $\boldsymbol{A}$ and 1 by $\boldsymbol{I}$. The series always converges.

Proof (slide 9) — differentiate term by term:

$$\frac{\mathrm{d}}{\mathrm{d}t}\exp(\boldsymbol{A}t) = \sum_{k=1}^{\infty} \frac{k\,t^{k-1}}{k\,(k-1)!}\boldsymbol{A}^k = \sum_{k=0}^{\infty} \frac{t^k}{k!}\boldsymbol{A}^{k+1} = \boldsymbol{A}\exp(\boldsymbol{A}t),$$

so $\dot{\boldsymbol{\Phi}} = \boldsymbol{A}\boldsymbol{\Phi}$; and at $t = 0$ only the $k = 0$ term survives: $\exp(0) = \boldsymbol{I}$, $\det \boldsymbol{I} = 1 \neq 0$. Both conditions of Definition 1.2 hold.

5.2 Useful properties

Property	Formula	Comment
start value	$e^{\boldsymbol{A}\cdot 0} = \boldsymbol{I}$	always check this first
derivative	$\frac{\mathrm{d}}{\mathrm{d}t}e^{\boldsymbol{A}t} = \boldsymbol{A}e^{\boldsymbol{A}t} = e^{\boldsymbol{A}t}\boldsymbol{A}$	at $t = 0$ this gives $\boldsymbol{A}$ (Task 2)
addition of times	$e^{\boldsymbol{A}(t+s)} = e^{\boldsymbol{A}t}e^{\boldsymbol{A}s}$	the composition property for LTI systems
inverse	$\big(e^{\boldsymbol{A}t}\big)^{-1} = e^{-\boldsymbol{A}t}$	backwards in time
diagonal $\boldsymbol{A}$	$e^{\mathrm{diag}(\lambda_i)t} = \mathrm{diag}\big(e^{\lambda_i t}\big)$	element by element **only** for diagonal matrices
block-diagonal	$e^{\mathrm{diag}(\boldsymbol{A}_1,\boldsymbol{A}_2)t} = \mathrm{diag}\big(e^{\boldsymbol{A}_1 t}, e^{\boldsymbol{A}_2 t}\big)$	used in Task 3
warning	$e^{(\boldsymbol{A}+\boldsymbol{B})t} \neq e^{\boldsymbol{A}t}e^{\boldsymbol{B}t}$ in general	equality only if $\boldsymbol{AB} = \boldsymbol{BA}$

Common mistake: $e^{\boldsymbol{A}t}$ is **not** the matrix of $e^{a_{ij}t}$. For $\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ the element-wise exponential would be $\begin{bmatrix} 1 & e^t \\ 1 & 1 \end{bmatrix}$, but the true one is $\begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$.

Computing the infinite series by hand is impractical, so the lecture gives four methods (Sections 6, 7, 9, 10).

---

6. Method 1 — series expansion and nilpotent matrices

A matrix is **nilpotent** if $\boldsymbol{A}^k = 0$ for some k. Then all later powers vanish too and the series **terminates**:

$$\boldsymbol{A}^2 = 0 \quad \Longrightarrow \quad e^{\boldsymbol{A}t} = \boldsymbol{I} + \boldsymbol{A}t.$$

Slide 14: $\boldsymbol{A} = \begin{bmatrix} 1 & -1/2 \\ 2 & -1 \end{bmatrix}$, $\boldsymbol{A}^2 = \begin{bmatrix} 1-1 & -\frac{1}{2}+\frac{1}{2} \\ 2-2 & -1+1 \end{bmatrix} = 0$, so

$$\boldsymbol{\Phi}(t, t_0) = \boldsymbol{I} + (t - t_0)\boldsymbol{A} = \begin{bmatrix} 1 + (t-t_0) & -\frac{1}{2}(t-t_0) \\ 2(t-t_0) & 1 - (t-t_0) \end{bmatrix}.$$

A nilpotent matrix has **all eigenvalues equal to 0** (if $\boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}$ then $0 = \boldsymbol{A}^k\boldsymbol{v} = \lambda^k\boldsymbol{v}$, so $\lambda = 0$). For non-nilpotent matrices, truncating the series gives only an approximation, useful for small t.

---

7. Method 2 — similarity transform (diagonalization)

7.1 Idea: change coordinates so the states decouple

If $\boldsymbol{A}$ were diagonal, every state would evolve on its own: $\dot{x}_i = \lambda_i x_i$, so $x_i(t) = e^{\lambda_i t}x_i(0)$ — n independent scalar problems. Most matrices are not diagonal, but many become diagonal **in the right coordinates**: the coordinates along the eigenvectors.

Two observations (slide 11):

1. If $\boldsymbol{A}' = \boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P}$ ($\boldsymbol{A}'$ and $\boldsymbol{A}$ are **similar**), then $\exp(\boldsymbol{A}'t) = \boldsymbol{P}^{-1}\exp(\boldsymbol{A}t)\boldsymbol{P}$. *Reason:* $(\boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P})^k = \boldsymbol{P}^{-1}\boldsymbol{A}^k\boldsymbol{P}$ (all inner $\boldsymbol{P}\boldsymbol{P}^{-1}$ cancel), so every term of the series transforms the same way.
2. If $\boldsymbol{A}$ is **diagonalizable**, i.e. $\boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P} = \boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$, then

$$\boxed{\exp(\boldsymbol{A}t) = \boldsymbol{P}\exp(\boldsymbol{\Lambda}t)\boldsymbol{P}^{-1} = \boldsymbol{P}\,\mathrm{diag}\left(e^{\lambda_1 t}, \dots, e^{\lambda_n t}\right)\boldsymbol{P}^{-1}.}$$

- $\boldsymbol{A}$ is diagonalizable if it has n linearly independent eigenvectors $\boldsymbol{u}_i$; then $\boldsymbol{P} = [\boldsymbol{u}_1 \ \cdots \ \boldsymbol{u}_n]$. **Distinct eigenvalues always give independent eigenvectors.**
- Matrices with repeated eigenvalues *may* not be diagonalizable (→ Jordan form, §9). Real symmetric matrices always are.

Reading the formula right to left: $\boldsymbol{P}^{-1}$ converts the initial state into eigen-coordinates, $e^{\boldsymbol{\Lambda}t}$ lets each eigen-coordinate evolve on its own, $\boldsymbol{P}$ converts back.

7.2 Recipe

1. Eigenvalues: solve $\det(\lambda\boldsymbol{I} - \boldsymbol{A}) = 0$.
2. Eigenvector for each λ_i: solve $(\lambda_i\boldsymbol{I} - \boldsymbol{A})\boldsymbol{u}_i = 0$.
3. $\boldsymbol{P} = [\boldsymbol{u}_1 \ \cdots \ \boldsymbol{u}_n]$, compute $\boldsymbol{P}^{-1}$.
4. Multiply $\boldsymbol{P}\,\mathrm{diag}(e^{\lambda_i t})\boldsymbol{P}^{-1}$.
5. **Check:** at $t = 0$ the result must be $\boldsymbol{I}$; its derivative at $t = 0$ must be $\boldsymbol{A}$.

7.3 Worked lecture example (slides 12–13)

$\boldsymbol{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: $\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = (\lambda - 2)^2 - 1 = (\lambda - 1)(\lambda - 3)$, so $\lambda_1 = 1$, $\lambda_2 = 3$.

- $\lambda_1 = 1$: $\lambda_1 \boldsymbol{I} - \boldsymbol{A} = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \Rightarrow u_1 + u_2 = 0 \Rightarrow \boldsymbol{u}_1 = [1, -1]^\top$.
- $\lambda_2 = 3$: $\lambda_2 \boldsymbol{I} - \boldsymbol{A} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \Rightarrow u_1 = u_2 \Rightarrow \boldsymbol{u}_2 = [1, 1]^\top$.

$$\boldsymbol{P} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad \boldsymbol{\Lambda} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}, \quad \boldsymbol{P}^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix},$$

$$\boldsymbol{\Phi}(t, t_0) = \boldsymbol{P} \begin{bmatrix} e^{t-t_0} & 0 \\ 0 & e^{3(t-t_0)} \end{bmatrix} \boldsymbol{P}^{-1} = \frac{1}{2} \begin{bmatrix} e^{t-t_0} + e^{3(t-t_0)} & -e^{t-t_0} + e^{3(t-t_0)} \\ -e^{t-t_0} + e^{3(t-t_0)} & e^{t-t_0} + e^{3(t-t_0)} \end{bmatrix}.$$

For $t_0 = 0$, $\boldsymbol{x}_0 = [1, 1]^\top$: $\boldsymbol{x}(t) = [e^{3t}, e^{3t}]^\top$ — the initial state is exactly the eigenvector $\boldsymbol{u}_2$, so only e^{3t} appears (a preview of modes, §8).

7.4 Complex eigenvalues (needed for Task 1)

If $\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = 0$ has complex roots $\lambda_{1,2} = \sigma \pm j\omega$, **the recipe works unchanged** — the eigenvectors and $\boldsymbol{P}$ are complex, but the final $e^{\boldsymbol{A}t}$ is real. Practical points:

- **Eigenvectors come in conjugate pairs:** if $\boldsymbol{u}_1$ belongs to λ_1, then $\boldsymbol{u}_2 = \bar{\boldsymbol{u}}_1$ belongs to $\lambda_2 = \bar{\lambda}_1$. Compute one, conjugate for the other.
- **Finding $\boldsymbol{u}_1$:** $\lambda_1 \boldsymbol{I} - \boldsymbol{A}$ is singular, so one row suffices — pick the simpler row.
- **Simplify with Euler:** the result contains $e^{\lambda_1 t}$ and $e^{\lambda_2 t}$ in conjugate pairs; group them as $e^{\sigma t} \cos \omega t$ and $e^{\sigma t} \sin \omega t$ (§2.1).

Shortcut for any real 2×2 matrix with eigenvalues $\sigma \pm j\omega$ ($\omega \neq 0$):

$$\boxed{e^{\boldsymbol{A}t} = e^{\sigma t} \left[\cos(\omega t)\, \boldsymbol{I} + \frac{\sin(\omega t)}{\omega} (\boldsymbol{A} - \sigma \boldsymbol{I}) \right]}$$

Check: at $t = 0$ it gives $\boldsymbol{I}$; its derivative at $t = 0$ is $\sigma \boldsymbol{I} + (\boldsymbol{A} - \sigma \boldsymbol{I}) = \boldsymbol{A}$ ✓. (It follows from Cayley–Hamilton, §10.) Use it to **verify** the diagonalization result. *Example (slide 24):* $\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, $\sigma = 0$, $\omega = 1$: $e^{\boldsymbol{A}t} = \cos t\, \boldsymbol{I} + \sin t\, \boldsymbol{A} = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$ — a rotation: the state circles the origin forever.

---

8. Modes

8.1 The modal decomposition (slide 15)

Write $\boldsymbol{P} = [\boldsymbol{p}_1 \ \cdots \ \boldsymbol{p}_n]$ (columns = eigenvectors) and $\boldsymbol{P}^{-1}$ with rows $\tilde{\boldsymbol{p}}_1, \ldots, \tilde{\boldsymbol{p}}_n$. Multiplying out $\boldsymbol{P} \exp(\boldsymbol{\Lambda} t) \boldsymbol{P}^{-1}$ column-times-row gives a **sum of n modes**:

$$\boxed{\exp(\boldsymbol{A}t) = \sum_{i=1}^{n} \boldsymbol{p}_i \tilde{\boldsymbol{p}}_i \, e^{\lambda_i t}}$$

Each $\boldsymbol{p}_i\tilde{\boldsymbol{p}}_i$ is an $n \times n$ matrix (column × row). **Each term is called a mode** of $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}$. In general the solution contains contributions of all modes.

Useful facts about the mode matrices $\boldsymbol{M}_i = \boldsymbol{p}_i\tilde{\boldsymbol{p}}_i$:

- $\tilde{\boldsymbol{p}}_i\boldsymbol{p}_i = 1$ and $\tilde{\boldsymbol{p}}_j\boldsymbol{p}_i = 0$ for $i \neq j$ (because $\boldsymbol{P}^{-1}\boldsymbol{P} = \boldsymbol{I}$).
- $\sum_i \boldsymbol{M}_i = \boldsymbol{I}$ (set $t = 0$) — a quick check.
- For a complex pair, $\boldsymbol{M}_2 = \bar{\boldsymbol{M}}_1$, so the two modes add up to a real term $2\,\mathrm{Re}\left(\boldsymbol{M}_1 e^{\lambda_1 t}\right)$.

8.2 Exciting a single mode (slides 16–17)

If the initial state lies along one eigenvector, $\boldsymbol{x}(0) = \alpha\boldsymbol{p}_i$, then only that mode appears:

$$\boldsymbol{x}(t) = \exp(\boldsymbol{A}t)\boldsymbol{x}(0) = \sum_k \boldsymbol{p}_k \underbrace{\tilde{\boldsymbol{p}}_k\boldsymbol{p}_i}_{1 \text{ if } k=i,\ 0 \text{ else}} \alpha\, e^{\lambda_k t} = \alpha\boldsymbol{p}_i e^{\lambda_i t}.$$

The state moves **along the straight line through $\boldsymbol{p}_i$**, growing or shrinking like $e^{\lambda_i t}$.

Lecture example: $\boldsymbol{A} = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$, $\lambda_1 = -1$, $\lambda_2 = 1$, $\boldsymbol{P} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$, $\boldsymbol{P}^{-1} = \begin{bmatrix} 1 & -1/2 \\ 0 & 1/2 \end{bmatrix}$:

$$\boldsymbol{\Phi}(t) = \begin{bmatrix} 1 & -1/2 \\ 0 & 0 \end{bmatrix} e^{-t} + \begin{bmatrix} 0 & 1/2 \\ 0 & 1 \end{bmatrix} e^{t}.$$

For $\boldsymbol{x}_0 = \alpha\boldsymbol{p}_1 = [\alpha, 0]^\top$: $\boldsymbol{x}(t) = [\alpha, 0]^\top e^{-t}$ — the unstable mode e^t is **not present**, even though the system is unstable. Any tiny error in $\boldsymbol{x}_0$ would excite it, though.

8.3 Which mode diverges?

The size of mode i is governed by $|e^{\lambda_i t}| = e^{\mathrm{Re}(\lambda_i)\, t}$:

$\mathrm{Re}\,\lambda_i$	$\mathrm{Im}\,\lambda_i$	Mode behaviour as $t \to \infty$
< 0	0	decays monotonically
< 0	$\neq 0$	decaying oscillation
$= 0$	0	constant
$= 0$	$\neq 0$	sustained oscillation (bounded)
> 0	any	**diverges** ⇒ instability

A mode introduces divergence exactly when the real part of its eigenvalue is positive. If all real parts are negative, every mode decays and $e^{\boldsymbol{A}t} \to 0$. (Repeated eigenvalues with $\mathrm{Re}\,\lambda = 0$ can produce terms like t that diverge slowly — §9.)

---

9. Method 3 — Jordan form (non-diagonalizable A)

9.1 When eigenvectors are missing

If an eigenvalue is repeated, there may be fewer independent eigenvectors than its multiplicity; then $\boldsymbol{A}$ is **not diagonalizable**. One can still transform to the **Jordan canonical form** $\boldsymbol{J} = \boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P}$ (slide 18): block-diagonal, with a diagonal part $\boldsymbol{J}_0$ and **Jordan blocks**

$$\boldsymbol{J}_i = \begin{bmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix} = \lambda \boldsymbol{I} + \boldsymbol{N}_i,$$

where $\boldsymbol{N}_i$ has ones on the superdiagonal and is **nilpotent** ($\boldsymbol{N}_i^{n_i} = 0$).

9.2 Exponential of a Jordan block (slide 19)

$\lambda \boldsymbol{I}$ and $\boldsymbol{N}_i$ commute, so $e^{\boldsymbol{J}_i t} = e^{\lambda t} e^{\boldsymbol{N}_i t}$, and the series for $e^{\boldsymbol{N}_i t}$ terminates:

$$e^{\boldsymbol{J}_i t} = e^{\lambda t} \begin{bmatrix} 1 & t & \cdots & \frac{t^{n_i-1}}{(n_i-1)!} \\ 0 & 1 & \cdots & \frac{t^{n_i-2}}{(n_i-2)!} \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}. \qquad \text{For a 2×2 block:} \quad \exp\!\left(\begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} t \right) = e^{\lambda t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

Consequence: non-diagonalizable matrices produce terms $t\,e^{\lambda t}$, $t^2 e^{\lambda t}$, ... in the response. For $\lambda < 0$ they still decay (the exponential wins over the polynomial), but they can first **rise** — e.g. $t\,e^{-t}$ grows to e^{-1} at $t = 1$ before decaying.

Example (slide 25): $\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is a Jordan block with $\lambda = 0$: $e^{\boldsymbol{A}t} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$, and $\boldsymbol{x}_0 = [\alpha, \beta]^\top$ gives $\boldsymbol{x}(t) = [\alpha + \beta t,\ \beta]^\top$ — a mass moving at constant speed β. Here the term t diverges even though Re $\lambda = 0$.

If $\boldsymbol{A}$ is already in Jordan form (as in Task 3), no transformation is needed — just exponentiate block by block.

---

10. Method 4 — Cayley–Hamilton theorem

10.1 The theorem

Every square matrix satisfies its own characteristic equation: if $p(\lambda) = \det(\lambda \boldsymbol{I} - \boldsymbol{A}) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0$, then

$$p(\boldsymbol{A}) = \boldsymbol{A}^n + c_{n-1}\boldsymbol{A}^{n-1} + \cdots + c_0 \boldsymbol{I} = 0.$$

Example: $\boldsymbol{A} = \begin{bmatrix} -6 & -1 \\ 5 & 0 \end{bmatrix}$, $p(\lambda) = \lambda^2 + 6\lambda + 5$. Then $\boldsymbol{A}^2 = \begin{bmatrix} 31 & 6 \\ -30 & -5 \end{bmatrix}$ and $\boldsymbol{A}^2 + 6\boldsymbol{A} + 5\boldsymbol{I} = \begin{bmatrix} 31 - 36 + 5 & 6 - 6 \\ -30 + 30 & -5 + 0 + 5 \end{bmatrix} = 0$ ✓.

Why this helps: $\boldsymbol{A}^n$ can be written in terms of $\boldsymbol{I}, \boldsymbol{A}, \dots, \boldsymbol{A}^{n-1}$; so can $\boldsymbol{A}^{n+1} = \boldsymbol{A}\cdot\boldsymbol{A}^n$, and so on. Hence **every power — and every power series, including $e^{\boldsymbol{A}t}$ — collapses to a polynomial of degree $n-1$** (slide 21):

$$\exp(\boldsymbol{A}t) = \alpha_0(t)\boldsymbol{I} + \alpha_1(t)\boldsymbol{A} + \cdots + \alpha_{n-1}(t)\boldsymbol{A}^{n-1}.$$

Only the n scalar functions $\alpha_k(t)$ are unknown. For a 2×2 matrix: $e^{\boldsymbol{A}t} = \alpha_0(t)\boldsymbol{I} + \alpha_1(t)\boldsymbol{A}$.

10.2 Finding the coefficients: distinct eigenvalues (slides 22–23)

Multiply the polynomial identity by an eigenvector $\boldsymbol{u}_i$: since $\boldsymbol{A}^k\boldsymbol{u}_i = \lambda_i^k\boldsymbol{u}_i$, both sides become multiples of $\boldsymbol{u}_i$, giving a **scalar** equation for every eigenvalue:

$$e^{\lambda_i t} = \sum_{k=0}^{n-1} \alpha_k(t)\lambda_i^k, \qquad i = 1, \dots, n.$$

In matrix form, with the **Vandermonde matrix $\boldsymbol{V}$**:

$$\begin{bmatrix} e^{\lambda_1 t} \\ \vdots \\ e^{\lambda_n t} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \lambda_1 & \cdots & \lambda_1^{n-1} \\ \vdots & & & \vdots \\ 1 & \lambda_n & \cdots & \lambda_n^{n-1} \end{bmatrix}}_{\boldsymbol{V}} \begin{bmatrix} \alpha_0(t) \\ \vdots \\ \alpha_{n-1}(t) \end{bmatrix}.$$

$\boldsymbol{V}$ is invertible **iff all eigenvalues are distinct**; then $[\alpha_0, \dots, \alpha_{n-1}]^\top = \boldsymbol{V}^{-1}[e^{\lambda_1 t}, \dots, e^{\lambda_n t}]^\top$. **No eigenvectors are needed** — only eigenvalues.

2×2 with distinct real eigenvalues $\lambda_1 \neq \lambda_2$ — solving the two equations $e^{\lambda_i t} = \alpha_0 + \alpha_1\lambda_i$:

$$\alpha_1(t) = \frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2}, \qquad \alpha_0(t) = \frac{\lambda_1 e^{\lambda_2 t} - \lambda_2 e^{\lambda_1 t}}{\lambda_1 - \lambda_2}.$$

(Subtract the two equations to get α_1; substitute back for α_0.)

10.3 Recipe

1. Characteristic polynomial and eigenvalues.
2. Write $e^{\boldsymbol{A}t} = \alpha_0\boldsymbol{I} + \alpha_1\boldsymbol{A} + \dots$ (degree $n-1$).
3. One equation $e^{\lambda_i t} = \sum \alpha_k\lambda_i^k$ per eigenvalue (derivative equations for repeated ones, §10.5).
4. Solve for the $\alpha_k(t)$.
5. Insert into the matrix polynomial and multiply out.
6. **Check** $t = 0 \Rightarrow \boldsymbol{I}$.

10.4 Complex eigenvalues (slide 24)

$\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, $\lambda^2 + 1 = 0$, $\lambda_{1,2} = \pm j$:

$e^{jt} = \cos t + j\sin t = \alpha_0 + j\alpha_1, \qquad e^{-jt} = \cos t - j\sin t = \alpha_0 - j\alpha_1 \implies \alpha_0 = \cos t,\ \alpha_1 = \sin t,$

(add / subtract the two equations), so $\boldsymbol{\Phi}(t) = \cos t\, \boldsymbol{I} + \sin t\, \boldsymbol{A} = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$. For general $\sigma \pm j\omega$ the same steps give the shortcut of §7.4.

10.5 Repeated eigenvalues (slides 26–28)

If λ_i has algebraic multiplicity $\mu_i \geq 2$, the Vandermonde matrix has identical rows and is singular. **Fix:** differentiate the scalar identity $e^{\lambda t} = \sum \alpha_k \lambda^k$ with respect to λ (the α_k do not depend on λ) to get $\mu_i - 1$ additional equations:

$$t\, e^{\lambda_i t} = \alpha_1 + 2\alpha_2 \lambda_i + \cdots + (n-1)\alpha_{n-1}\lambda_i^{n-2}, \qquad t^2 e^{\lambda_i t} = 2\alpha_2 + \ldots, \quad \ldots$$

These replace the duplicate rows.

Lecture example: $\boldsymbol{A} = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \\ -1 & 0 & 3 \end{bmatrix}$, $\lambda_1 = 2$, $\lambda_{2,3} = 1$. Ansatz $e^{\lambda t} = \alpha_0 + \alpha_1 \lambda + \alpha_2 \lambda^2$:

$$\begin{aligned} \lambda = 1: \quad & e^t = \alpha_0 + \alpha_1 + \alpha_2 && && \alpha_0 = -2te^t + e^{2t} \\ \tfrac{\mathrm{d}}{\mathrm{d}\lambda} \text{ at } \lambda = 1: \quad & te^t = \alpha_1 + 2\alpha_2 && \Longrightarrow \quad && \alpha_1 = (2+3t)e^t - 2e^{2t} \\ \lambda = 2: \quad & e^{2t} = \alpha_0 + 2\alpha_1 + 4\alpha_2 && && \alpha_2 = -(t+1)e^t + e^{2t}. \end{aligned}$$

Then $\boldsymbol{\Phi}(t) = \alpha_0 \boldsymbol{I} + \alpha_1 \boldsymbol{A} + \alpha_2 \boldsymbol{A}^2$ with $\boldsymbol{A}^2 = \begin{bmatrix} -2 & 0 & 6 \\ 0 & 1 & 0 \\ -3 & 0 & 7 \end{bmatrix}$:

$$\boldsymbol{\Phi}(t) = \begin{bmatrix} 2e^t - e^{2t} & 0 & 2e^{2t} - 2e^t \\ 0 & e^t & 0 \\ e^t - e^{2t} & 0 & 2e^{2t} - e^t \end{bmatrix}.$$

Note: the α_k on slide 28 are correct, but the matrix printed there contains errors in most entries (e.g. terms like $-12t$, which cannot appear because the t-terms cancel). The matrix above is the correct one: it satisfies $\boldsymbol{\Phi}(0) = \boldsymbol{I}$ and $\dot{\boldsymbol{\Phi}} = \boldsymbol{A}\boldsymbol{\Phi}$, and agrees with a direct computation (`python/lecture_examples.py`). That the $t\, e^t$ terms cancel means this $\boldsymbol{A}$ is actually diagonalizable (the repeated eigenvalue 1 has two independent eigenvectors).

---

11. Which method to choose?

Method	Main idea	When is it useful?
Series expansion	evaluate $\sum t^k \boldsymbol{A}^k / k!$	nilpotent $\boldsymbol{A}$ ($\boldsymbol{A}^k = 0$) — the series terminates exactly; otherwise a small-t approximation
Diagonalization	transform to $\boldsymbol{\Lambda}$, exponentiate the diagonal	n independent, easy eigenvectors; **also gives the modes** (Task 1)
Jordan form	exponentiate each Jordan block via its nilpotent part	$\boldsymbol{A}$ not diagonalizable, or already in Jordan form (Task 3)

Method	Main idea	When is it useful?
Cayley–Hamilton	$e^{\boldsymbol{A}t}$ as a polynomial of degree $n-1$	small matrices, eigenvalues known but eigenvectors inconvenient (Task 4); repeated eigenvalues via derivatives

Whatever method you use, always run the two **checks**: $\boldsymbol{\Phi}(0) = \boldsymbol{I}$ and $\dot{\boldsymbol{\Phi}}(0) = \boldsymbol{A}$.

---

12. Response with an input: the convolution integral

12.1 Variation of parameters (slides 31–32)

Homogeneous part. $\boldsymbol{x}_h(t) = \exp[\boldsymbol{A}(t-t_0)]\boldsymbol{x}_0$ solves $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}$ (§5.1).

Particular part. Guess a solution of the form "homogeneous solution with a varying coefficient":

$$\boldsymbol{x}_p(t) = \exp(\boldsymbol{A}t)\,\boldsymbol{p}(t), \qquad \boldsymbol{p}(t) \text{ unknown.}$$

Insert into $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{r}$ (product rule on the left):

$$\boldsymbol{A}\exp(\boldsymbol{A}t)\boldsymbol{p}(t) + \exp(\boldsymbol{A}t)\dot{\boldsymbol{p}}(t) = \boldsymbol{A}\exp(\boldsymbol{A}t)\boldsymbol{p}(t) + \boldsymbol{r}(t) \implies \dot{\boldsymbol{p}}(t) = \exp(-\boldsymbol{A}t)\,\boldsymbol{r}(t).$$

Integrate with $\boldsymbol{p}(t_0) = 0$ (the initial condition is already in $\boldsymbol{x}_h$):

$$\boldsymbol{p}(t) = \int_{t_0}^{t} \exp(-\boldsymbol{A}\tau)\boldsymbol{r}(\tau)\,\mathrm{d}\tau \implies \boldsymbol{x}_p(t) = \int_{t_0}^{t} \exp\big(\boldsymbol{A}(t-\tau)\big)\boldsymbol{r}(\tau)\,\mathrm{d}\tau = \int_{t_0}^{t} \boldsymbol{\Phi}(t,\tau)\boldsymbol{r}(\tau)\,\mathrm{d}\tau.$$

This is the matrix version of the integrating-factor result in §1.2.

12.2 Total response (slide 33)

With $\boldsymbol{r}(t) = \boldsymbol{B}\boldsymbol{u}(t)$ and the output equation $\boldsymbol{y} = \boldsymbol{C}\boldsymbol{x} + \boldsymbol{D}\boldsymbol{u}$:

$$\boxed{\boldsymbol{x}(t, t_0, \boldsymbol{x}_0) = \boldsymbol{\Phi}(t, t_0)\boldsymbol{x}_0 + \int_{t_0}^{t} \boldsymbol{\Phi}(t,\tau)\boldsymbol{B}\boldsymbol{u}(\tau)\,\mathrm{d}\tau}$$

$$\boxed{\boldsymbol{y}(t) = \underbrace{\boldsymbol{C}\boldsymbol{\Phi}(t, t_0)\boldsymbol{x}_0}_{\text{zero-input / natural response}} + \underbrace{\boldsymbol{C}\int_{t_0}^{t} \boldsymbol{\Phi}(t,\tau)\boldsymbol{B}\boldsymbol{u}(\tau)\,\mathrm{d}\tau + \boldsymbol{D}\boldsymbol{u}(t)}_{\text{zero-state / forced response}}}$$

with $\boldsymbol{\Phi}(t,\tau) = e^{\boldsymbol{A}(t-\tau)}$ for LTI systems. $\boldsymbol{B}$ represents the **actuator structure** — how the input can act on the states.

12.3 Reading the convolution integral

Exactly as in the scalar case: the input at time τ gives a kick $\boldsymbol{B}\boldsymbol{u}(\tau)\,\mathrm{d}\tau$ to the state; by time t that kick has been propagated by $e^{\boldsymbol{A}(t-\tau)}$; the integral sums all past kicks. Only inputs **up to time** t enter — the response is **causal** (Week 2).

12.4 Change of variables (slides 34–35)

Substituting $\theta = t - \tau$ ($\tau = t - \theta$, $\mathrm{d}\tau = -\mathrm{d}\theta$, limits $\tau = 0 \mapsto \theta = t$, $\tau = t \mapsto \theta = 0$) gives the equivalent form

$$\boldsymbol{x}_p(t) = \int_0^t e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\boldsymbol{u}(\tau)\,\mathrm{d}\tau = \int_0^t e^{\boldsymbol{A}\theta}\boldsymbol{B}\boldsymbol{u}(t-\theta)\,\mathrm{d}\theta.$$

θ is "how long ago" — the second form weights the input from θ seconds ago by $e^{\boldsymbol{A}\theta}$. Use whichever form makes the integral easier.

---

13. Standard test signals, impulse and step response

13.1 Test signals (slide 36)

To compare systems, engineers apply standard inputs: the **unit step** $\sigma(t)$ (switch something on and keep it on), the **rectangle** $\Pi(t)$ and the **Dirac impulse** $\delta(t) = \lim_{\epsilon\to 0}\Pi(t)$ (a short hit). See §2.4.

13.2 Impulse response (slides 37–38)

Consider only the zero-state response and write the feedthrough also as an integral, $\boldsymbol{D}\boldsymbol{u}(t) = \int \boldsymbol{D}\boldsymbol{u}(\tau)\delta(t-\tau)\,\mathrm{d}\tau$ (sifting property):

$$\boldsymbol{y}(t) = \int_{t_0}^t \left[\boldsymbol{C}\boldsymbol{\Phi}(t,\tau)\boldsymbol{B} + \boldsymbol{D}\delta(t-\tau)\right]\boldsymbol{u}(\tau)\,\mathrm{d}\tau = \int_{t_0}^t \boldsymbol{H}(t,\tau)\boldsymbol{u}(\tau)\,\mathrm{d}\tau.$$

Definition 1.3 — impulse response matrix:

$$\boldsymbol{H}(t,\tau) = \begin{cases} \boldsymbol{C}\boldsymbol{\Phi}(t,\tau)\boldsymbol{B} + \boldsymbol{D}\delta(t-\tau), & t \geq \tau \\ 0, & t < \tau \end{cases}$$

For LTI systems it depends only on $t - \tau$, so with $\tau = 0$:

$$\boxed{\boldsymbol{H}(t) = \begin{cases} \boldsymbol{C}e^{\boldsymbol{A}t}\boldsymbol{B} + \boldsymbol{D}\delta(t), & t \geq 0 \\ 0, & t < 0 \end{cases}} \qquad \text{and} \qquad \boldsymbol{y}(t) = \boldsymbol{C}e^{\boldsymbol{A}(t-t_0)}\boldsymbol{x}_0 + \boldsymbol{H}(t) * \boldsymbol{u}(t).$$

Why it is called "impulse response": apply $u(t) = \delta(t)$ with $\boldsymbol{x}(0) = 0$. The sifting property collapses the integral to the value of the integrand at $\tau = 0$:

$$\boldsymbol{x}(t) = \int_{0^-}^t e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\,\delta(\tau)\,\mathrm{d}\tau = e^{\boldsymbol{A}t}\boldsymbol{B}, \qquad \boldsymbol{y}(t) = \boldsymbol{C}e^{\boldsymbol{A}t}\boldsymbol{B} + \boldsymbol{D}\delta(t) = \boldsymbol{H}(t) \qquad (t \geq 0).$$

Intuition: the impulse instantly sets the state to $\boldsymbol{B}$ ("the hammer blow puts the state there"), after which the system evolves freely: $\boldsymbol{x}(t) = e^{\boldsymbol{A}t}\boldsymbol{B}$. So **an impulse response is a natural response started from** $\boldsymbol{x}(0^+) = \boldsymbol{B}$. The entry $h_{ij}(t)$ is a dynamic input-to-output sensitivity: how much output i depends on what input j was t seconds ago (slide 41). Knowing $\boldsymbol{H}(t)$ gives the response to *any* input via the convolution — it is the system's fingerprint.

13.3 Step response — and steps that start at t_0 (needed for Task 4)

For a constant input $\boldsymbol{u}(\tau) = \boldsymbol{u}_c$ on $[t_0, t]$ and invertible $\boldsymbol{A}$:

$$\int_{t_0}^{t} e^{\boldsymbol{A}(t-\tau)}\,\mathrm{d}\tau = \boldsymbol{A}^{-1}\big(e^{\boldsymbol{A}(t-t_0)} - \boldsymbol{I}\big) \qquad \Longrightarrow \qquad \boldsymbol{x}_p(t) = \boldsymbol{A}^{-1}\big(e^{\boldsymbol{A}(t-t_0)} - \boldsymbol{I}\big)\boldsymbol{B}\boldsymbol{u}_c.$$

Derivation: with $\theta = t - \tau$ the integral is $\int_0^{t-t_0} e^{\boldsymbol{A}\theta}\,\mathrm{d}\theta$; since $\frac{\mathrm{d}}{\mathrm{d}\theta}\big(\boldsymbol{A}^{-1}e^{\boldsymbol{A}\theta}\big) = e^{\boldsymbol{A}\theta}$, the antiderivative is $\boldsymbol{A}^{-1}e^{\boldsymbol{A}\theta}$, evaluated between 0 and $t - t_0$. (Matrix version of §1.3; $\boldsymbol{A}^{-1}$ and $e^{\boldsymbol{A}\theta}$ commute.)

Steady state (all eigenvalues with $\mathrm{Re}\,\lambda < 0$, so $e^{\boldsymbol{A}(t-t_0)} \to 0$):

$$\boldsymbol{x}(\infty) = -\boldsymbol{A}^{-1}\boldsymbol{B}\boldsymbol{u}_c, \qquad \boldsymbol{y}(\infty) = \big(-\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B} + \boldsymbol{D}\big)\boldsymbol{u}_c.$$

Sanity check: at steady state $\dot{\boldsymbol{x}} = 0 = \boldsymbol{A}\boldsymbol{x}(\infty) + \boldsymbol{B}\boldsymbol{u}_c$ gives the same $\boldsymbol{x}(\infty)$ ✓.

Delayed step $u(t) = \sigma(t - t_0)$ **with the state given at** t_0**.** Split the time axis:

- $t \le t_0$**:** the input is zero, so only the natural motion acts. Since the state is known at t_0, use the state transition matrix **backwards** (§4.2): $\boldsymbol{x}(t) = \boldsymbol{\Phi}(t, t_0)\boldsymbol{x}(t_0) = e^{\boldsymbol{A}(t-t_0)}\boldsymbol{x}(t_0)$ with $t - t_0 \le 0$.
- $t \ge t_0$**:** natural response from $\boldsymbol{x}(t_0)$ plus forced response of a constant input switched on at t_0: $\boldsymbol{x}(t) = e^{\boldsymbol{A}(t-t_0)}\boldsymbol{x}(t_0) + \boldsymbol{A}^{-1}\big(e^{\boldsymbol{A}(t-t_0)} - \boldsymbol{I}\big)\boldsymbol{B}$.
- Both expressions give $\boldsymbol{x}(t_0)$ at $t = t_0$ — the state (and, with $\boldsymbol{D} = 0$, the output) is **continuous**.

Time invariance pays off: the result depends only on $t - t_0$. The response to a step at t_0 is the response to a step at 0, shifted by t_0 (Week 2, Definition 1.2).

---

14. Worked lecture example: total response

Slides 39–41: $\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\boldsymbol{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\boldsymbol{C} = \begin{bmatrix} 0 & 1 \end{bmatrix}$, $\boldsymbol{D} = 0$, $\boldsymbol{x}_0 = [1, -1]^\top$, $t_0 = 0$, $u(t) = \sigma(t)$. (A mass: x_1 position, x_2 velocity, input = force, output = velocity.)

1. **State transition matrix.** $\boldsymbol{A}$ is nilpotent, so $\boldsymbol{\Phi}(t) = \boldsymbol{I} + t\boldsymbol{A} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$.
2. **Natural response.** $y_h(t) = \boldsymbol{C}\boldsymbol{\Phi}(t)\boldsymbol{x}_0 = \begin{bmatrix} 0 & 1 \end{bmatrix}\begin{bmatrix} 1 - t \\ -1 \end{bmatrix} = -1$ — the velocity stays -1 without a force.
3. **Forced response.** $y_p(t) = \boldsymbol{C}\int_0^t \boldsymbol{\Phi}(\tau)\boldsymbol{B}\,\mathrm{d}\tau = \begin{bmatrix} 0 & 1 \end{bmatrix}\begin{bmatrix} \int_0^t \tau\,\mathrm{d}\tau \\ \int_0^t 1\,\mathrm{d}\tau \end{bmatrix} = \begin{bmatrix} 0 & 1 \end{bmatrix}\begin{bmatrix} t^2/2 \\ t \end{bmatrix} = t$ — a unit force accelerates the mass at 1 m/s². (The slide uses the θ-form of §12.4.)
4. **Total response.** $y(t) = y_h + y_p = t - 1$.

5. **Impulse response.** $H(t) = \boldsymbol{C}\boldsymbol{\Phi}(t)\boldsymbol{B} = \begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}t\\1\end{bmatrix} = 1$: a hammer blow gives the mass a constant velocity forever — the sensitivity is constant in time (special case).

---

15. Two "inverse" problems from the exercise

Usually we are given $\boldsymbol{A}$ and $\boldsymbol{x}_0$ and compute the response. Tasks 2 and 3 go the other way.

15.1 Given trajectories → find $\boldsymbol{A}$ and $\boldsymbol{\Phi}$ (Task 2)

1. Read off the initial states $\boldsymbol{x}^{(i)}(0)$ by setting $t = 0$ in each trajectory (check they match the given ones).
2. $\boldsymbol{X}(t) = [\boldsymbol{x}^{(1)}(t) \ \cdots \ \boldsymbol{x}^{(n)}(t)]$, $\boldsymbol{X}_0 = \boldsymbol{X}(0)$.
3. $\boldsymbol{\Phi}(t,0) = \boldsymbol{X}(t)\boldsymbol{X}_0^{-1}$ (§4.3).
4. $\boldsymbol{A} = \dot{\boldsymbol{\Phi}}(0)$.
5. Checks: $\boldsymbol{\Phi}(0) = \boldsymbol{I}$, $\dot{\boldsymbol{\Phi}} = \boldsymbol{A}\boldsymbol{\Phi}$; each given trajectory satisfies $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}$.

Bonus insight: the exponentials appearing in the trajectories are the modes $e^{\lambda_i t}$, so the eigenvalues can be read off directly. A trajectory containing only **one** exponential started on an eigenvector (§8.2).

15.2 Given a desired natural response → find $\boldsymbol{x}_0$ (Task 3)

1. Compute $e^{\boldsymbol{A}t}$ (any method).
2. Write $\boldsymbol{x}_0 = [a, b, c, \dots]^\top$ with unknowns and compute $y(t) = \boldsymbol{C}e^{\boldsymbol{A}t}\boldsymbol{x}_0$ — a combination of $e^{\lambda_i t}$, $te^{\lambda_i t}$, ... with coefficients linear in a, b, c.
3. **Match coefficients** with the desired $y(t)$ (§2.3): one linear equation per function.
4. Solve; check by substitution.

What is possible? Only functions that appear as modes of $\boldsymbol{A}$ (and are seen through $\boldsymbol{C}$) can be produced. $y(t) = te^{-t}$ is possible only because $\boldsymbol{A}$ has a 2×2 Jordan block for $\lambda = -1$; a system with only simple eigenvalues could never produce a $t\,e^{\lambda t}$ term. And to keep the unstable mode out of $y(t)$, the initial state must **not excite it**.

---

16. Cheat sheet

State transition matrix

$$\dot{\boldsymbol{\Phi}} = \boldsymbol{A}\boldsymbol{\Phi},\ \boldsymbol{\Phi}(t_0,t_0) = \boldsymbol{I}, \qquad \boldsymbol{\Phi}(t,t_0) = e^{\boldsymbol{A}(t-t_0)}\,(\mathsf{LTI}), \qquad \boldsymbol{\Phi}(t,\tau) = \boldsymbol{\Phi}(t,\sigma)\boldsymbol{\Phi}(\sigma,\tau), \qquad \boldsymbol{\Phi}(t,t_0)^{-1} = \boldsymbol{\Phi}(t_0$$

Computing $e^{\boldsymbol{A}t}$

Situation	Formula
nilpotent ($\boldsymbol{A}^2 = 0$)	$\boldsymbol{I} + \boldsymbol{A}t$
diagonalizable	$\boldsymbol{P}\,\mathrm{diag}(e^{\lambda_i t})\boldsymbol{P}^{-1} = \sum_i \boldsymbol{p}_i\tilde{\boldsymbol{p}}_i e^{\lambda_i t}$
2×2, $\lambda = \sigma \pm j\omega$	$e^{\sigma t}\big[\cos\omega t\,\boldsymbol{I} + \frac{\sin\omega t}{\omega}(\boldsymbol{A} - \sigma\boldsymbol{I})\big]$

Situation	Formula
Jordan block $\begin{bmatrix}\lambda & 1\\ 0 & \lambda\end{bmatrix}$	$e^{\lambda t}\begin{bmatrix}1 & t\\ 0 & 1\end{bmatrix}$
Cayley–Hamilton, 2×2, $\lambda_1 \neq \lambda_2$	$\alpha_0\boldsymbol{I} + \alpha_1\boldsymbol{A}$, $\alpha_1 = \frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2}$, $\alpha_0 = \frac{\lambda_1 e^{\lambda_2 t} - \lambda_2 e^{\lambda_1 t}}{\lambda_1 - \lambda_2}$
repeated eigenvalue (CH)	add $\frac{\mathrm{d}}{\mathrm{d}\lambda}$: $te^{\lambda t} = \alpha_1 + 2\alpha_2\lambda + \ldots$

Always check $e^{\boldsymbol{A}\cdot 0} = \boldsymbol{I}$ and $\frac{\mathrm{d}}{\mathrm{d}t}e^{\boldsymbol{A}t}\big|_0 = \boldsymbol{A}$.

Response

$$\boldsymbol{y}(t) = \boldsymbol{C}e^{\boldsymbol{A}(t-t_0)}\boldsymbol{x}_0 + \boldsymbol{C}\int_{t_0}^{t} e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\boldsymbol{u}(\tau)\,\mathrm{d}\tau + \boldsymbol{D}\boldsymbol{u}(t)$$

- Impulse ($\boldsymbol{x}_0 = 0$): $\boldsymbol{x}(t) = e^{\boldsymbol{A}t}\boldsymbol{B}$, $\boldsymbol{H}(t) = \boldsymbol{C}e^{\boldsymbol{A}t}\boldsymbol{B} + \boldsymbol{D}\delta(t)$.
- Step from t_0: $\boldsymbol{x}_p = \boldsymbol{A}^{-1}(e^{\boldsymbol{A}(t-t_0)} - \boldsymbol{I})\boldsymbol{B}$; steady state $\boldsymbol{y}(\infty) = -\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B} + \boldsymbol{D}$.
- State known at t_0, earlier times: $\boldsymbol{x}(t) = e^{\boldsymbol{A}(t-t_0)}\boldsymbol{x}(t_0)$, $t < t_0$.

Modes: mode i diverges $\Leftrightarrow \operatorname{Re}\lambda_i > 0$. $\boldsymbol{x}_0 = \alpha\boldsymbol{p}_i$ excites only mode i. Trajectories from data: $\boldsymbol{\Phi} = \boldsymbol{X}(t)\boldsymbol{X}_0^{-1}$, $\boldsymbol{A} = \dot{\boldsymbol{\Phi}}(0)$.